

Spring 2024 Exam 1

Table 1 – Monthly Returns. Show your answers as percentages. Round to two Decimals. (40 points)

Month	Raw Returns		Value-Weighted Index Adjusted		S&P 500 Index Adjusted	
	Average	Median	Average	Median	Average	Median
1	3.26%	1.99%	1.17%	-1.14%	1.48%	-0.86%
2	2.05%	3.09%	0.94%	4.14%	0.81%	4.38%
3	0.49%	-1.36%	-1.37%	-2.94%	-1.61%	-3.40%
4	8.20%	7.29%	6.69%	5.65%	6.50%	5.67%
5	-16.01%	-11.73%	-13.57%	-11.20%	-14.18%	-11.99%
6	-13.62%	-13.39%	-9.60%	-9.36%	-9.42%	-9.06%
7	-0.92%	0.88%	3.68%	3.64%	3.63%	3.47%
8	9.43%	10.26%	8.83%	7.40%	9.46%	7.82%
9	2.05%	4.72%	-4.01%	-1.07%	-3.81%	-0.96%
10	9.76%	9.64%	8.70%	7.26%	8.57%	7.23%
11	1.65%	1.25%	-2.31%	-2.98%	-2.47%	-3.17%
12	-8.21%	-4.45%	-10.71%	-7.09%	-10.84%	-7.19%
13	4.59%	2.57%	7.01%	4.85%	7.20%	5.14%
14	7.69%	5.31%	4.31%	0.48%	4.08%	0.28%
15	-16.09%	-6.60%	-18.81%	-8.55%	-18.84%	-8.51%
16	3.97%	6.35%	5.76%	8.96%	5.68%	8.92%
17	-1.61%	-3.16%	-2.39%	-0.25%	-2.44%	-0.45%
18	3.66%	4.75%	1.73%	5.25%	1.55%	4.94%
19	1.57%	8.29%	-1.31%	4.01%	-1.34%	3.79%
20	-1.95%	-0.48%	-5.05%	-4.35%	-4.95%	-4.27%
21	1.16%	3.42%	-1.89%	-0.55%	-2.01%	-0.47%
22	-14.19%	-1.71%	-8.22%	-3.85%	-8.11%	-3.68%
23	-7.32%	-5.70%	11.47%	13.44%	10.06%	12.04%
24	19.99%	25.18%	6.83%	5.71%	7.07%	5.88%
25	11.95%	21.59%	3.84%	13.83%	4.87%	14.84%
26	0.55%	2.40%	-2.75%	0.74%	-2.19%	1.03%
27	10.46%	14.40%	5.03%	10.67%	5.25%	10.89%
28	-2.55%	-3.97%	-5.54%	-5.69%	-5.58%	-5.51%
29	11.06%	5.63%	10.95%	5.63%	11.57%	5.75%
30	-4.75%	-6.11%	-8.47%	-10.41%	-8.23%	-9.87%
31	12.62%	11.73%	6.99%	6.99%	8.02%	8.01%
32	18.41%	-1.29%	14.37%	-4.97%	15.90%	-4.03%
33	14.68%	12.37%	11.27%	8.90%	12.16%	9.99%
34	-0.65%	-1.78%	0.40%	0.60%	-0.29%	0.38%
35	2.88%	2.77%	-2.71%	-2.67%	-3.35%	-3.37%
36	-4.20%	-3.05%	-5.19%	-5.19%	-5.64%	-5.74%

Table 2 – Buy and Hold Returns Over 36**Months. Show your answers as percentages. Round to two Decimals (40 points).**

Month	Raw Returns		Value-Weighted Index Adjusted		S&P 500 Index Adjusted	
	Average	Median	Average	Median	Average	Median
1	3.258%	1.992%	1.175%	-1.141%	1.482%	-0.858%
2	5.765%	5.145%	2.591%	2.277%	2.796%	2.751%
3	9.799%	-4.896%	4.375%	-10.997%	4.345%	-11.172%
4	23.846%	2.040%	16.598%	-6.224%	16.354%	-6.345%
5	2.840%	-4.645%	-1.648%	-13.328%	-2.539%	-14.108%
6	-7.674%	-21.945%	-6.943%	-23.777%	-7.627%	-24.480%
7	-3.301%	-17.599%	1.352%	-21.173%	0.755%	-22.005%
8	-0.053%	-13.313%	4.673%	-8.532%	4.674%	-8.241%
9	10.651%	-3.559%	8.381%	-5.983%	8.742%	-5.766%
10	17.533%	2.374%	14.210%	-0.618%	14.458%	-0.261%
11	17.714%	2.040%	9.843%	-6.411%	9.924%	-6.309%
12	17.888%	2.542%	10.166%	-5.743%	10.235%	-5.525%
13	13.137%	0.285%	8.099%	-7.809%	8.405%	-7.526%
14	17.674%	7.890%	9.734%	-0.113%	9.844%	-0.083%
15	18.119%	12.238%	7.087%	0.036%	7.134%	-0.007%
16	21.940%	16.748%	13.340%	10.121%	13.306%	10.220%
17	25.565%	33.713%	15.782%	21.692%	15.675%	21.761%
18	20.542%	41.318%	8.475%	29.117%	8.155%	28.817%
19	28.026%	58.366%	12.989%	42.134%	12.614%	41.609%
20	31.280%	47.251%	12.425%	28.128%	12.123%	27.798%
21	43.165%	45.580%	20.397%	21.415%	19.930%	20.997%
22	38.168%	46.959%	22.846%	20.523%	22.507%	20.262%
23	1.686%	26.610%	9.994%	38.449%	7.964%	36.459%
24	28.465%	33.211%	24.087%	29.520%	22.547%	27.579%
25	31.113%	29.869%	18.547%	21.326%	18.082%	20.503%
26	43.027%	42.985%	26.788%	26.616%	26.928%	26.775%
27	46.534%	39.114%	24.591%	15.886%	25.034%	16.333%
28	57.316%	34.871%	31.593%	9.304%	31.989%	9.238%
29	66.208%	30.297%	40.406%	5.621%	41.539%	7.084%
30	71.824%	25.435%	41.771%	-2.199%	43.317%	-0.802%
31	120.057%	20.329%	82.365%	-19.368%	85.432%	-15.280%
32	143.721%	15.025%	100.189%	-28.072%	105.707%	-22.382%
33	174.812%	9.567%	125.474%	-40.149%	132.597%	-32.924%
34	175.878%	1.325%	127.125%	-47.779%	133.276%	-42.035%
35	156.622%	-4.323%	99.413%	-60.787%	104.788%	-55.912%
36	181.504%	-9.974%	121.673%	-67.019%	126.452%	-62.328%

Table 3: Buy and Hold Return Distribution. Show your answers as percentages. Round to two Decimals (40 points)

Percentile	Three Year HPR
Average	20.98% Evo Payments Inc
Smallest	-76.67% GreenSky Inc
25th	-63.28% Evelo Biosciences
50th	20.98% Evo Payments Inc
75th	24.98% Construction Partners
Largest	242.6% Inspire Medical Systems

**Note there were only 5 companies of the 10 that were in the SAS file list so Average and 50th Percentile are the same company and percentage return.*

Discussion Question 1 (40 points). Examine the most underperforming IPO in your sample. Is the underperformance driven by a sudden shock(s) or is it persistent throughout the sample period? Describe the contributing factors for this underperformance in one or two paragraphs. Cite your sources.

I found that the reason why GreenSky Inc.'s stock (GSKY) was producing negative returns was that its stocks were overvalued from its IPO. It was stated under Goldman Sachs "Media Relations" page that the acquisition of GreenSky Inc, the fintech platform for consumer loans, was an all-stock transaction.¹

"When an acquirer's shares are considered overvalued by the market relative to the target company's shares, stock financing is more appropriate."²

The IPO also came during a FinTech boom. After acquisition Goldman Sachs struggled to get GreenSky Inc back to where it once was, referring to the six years of profitability before going public.

Discussion Question 2 (40 points). Examine the most overperforming IPO in your sample. Is the overperformance driven by a sudden shock(s) or is it persistent throughout the sample period? Describe the contributing factors for this overperformance in one or two paragraphs. Cite your sources.

The Inspire Medical Systems was acquired by Medical Alley Association in May of 2018. There are several factors that contributed to the success of, not just their IPO, but consistent success in returns with a steady upward trend up through the 36 months of this project study.

The Medical Alley community has proven fertile ground for developing treatments for the most challenging diseases and conditions. It was the largest financing within the portfolio of companies acquired by Medical Alley in 2018.³

They are in a very specific sector of the medical industry. Along with this they are capturing most of the sleep apnea equipment sales being, as stated on their website, “the first and only FDA approved obstructive sleep apnea device”.⁴ They are the leading and, in some cases, the only option for medical providers.

Citations:

- 1) Goldman Sachs (2021, September 15). *Goldman Sachs Acquire GreenSky*. Goldmansachs.com. Retrieved March 15, 2024, from <https://www.goldmansachs.com/media-relations/press-releases/2021/goldman-sachs-to-acquire-greensky.html>
- 2) Chang, R. P., & Moore, K. M. *Corporate Finance* (2nd ed., p. 421). Pearson.
- 3) Medical Alley (2018, May 9). *Congratulations to Inspire Medical on their Successful IPO - Medical Alley*. Medicalalley.com. Retrieved March 15, 2024, from <https://medicalalley.org/congratulations-to-inspire-medical-on-their-successful-ipo/#:~:text=The%20Medical%20Alley%20Association%20congratulates,2018%20and%20the%20first%20IPO.>
- 4) Nelson, G. (n.d.). *Inspire*. Inspiresleep.com. Retrieved March 15, 2024, from <https://www.inspiresleep.com/en-us/about-inspire/>